<u>CMPT 276 Phase 1 Statement, Group 8</u>

<u>Game Objective(s):</u> Collects all “rewards” and evacuate the maze.

<u>Game Overview</u>

- The player **collects all “rewards” and reach the “exit”** to complete a **run**.
- The game contains **three maps** of **increasing layout complexities.**
- In each map, the player can complete multiple runs; after each run, the **player respawns on the same map and the difficulty level increases.**
- **Core loop: turn-based**. The player takes one move, then **all enemies** each take one move. Each “tick” is represented by a “turn.”

<u>Game Mechanisms</u>

- Reward mechanism:
  - The player can get extra score by methods including **collecting bonus rewards** or **killing enemies**.
  - The player should also be rewarded extra score **inversely (proportional) to the number of steps used.**
- Penalty mechanism:
  - The player gets penalized upon **encountering an enemy** or **triggering a trap**.
  - The penalty severity may vary, such as **score deduction, respawn,** or **game over.**
- Difficulty mechanism:
  - When the player collects all “rewards” and reaches “exit,” the game advances and the difficulty level increases.
  - Difficulty increases are implemented by:
    - Increasing the **number of enemies, traps, and/or rewards.**
    - (**Optional) increasing enemy speed/aggressiveness**
- Kill mechanism:
  - The player can **pick up “landmines”** scattered randomly throughout the map.
  - The player can then activate and place it throughout the maze, where **an enemy is killed if it stepped on an activated landmine**.
  - The **player should also be penalized** if stepped on an activated landmine.

<u>Overall Plan</u>

- The game will be implemented in **Java with Maven,** and version control via **Git/Github**.
- **Goal 1:** Map representation, player movement, turn-based loop, and level completion.
- **Goal 2:** Enemy, trap/landmine, and reward logic, encounter detection, and penalty handling.
- **Goal 3:** Scoring logic, level/difficulty scaling, and final testing/polishing.
- **Workflow:** Use feature branches and pull requests; merge after review and testing.